LIBBY — PRECLINICAL HORIZON

ulms-highgrade-pulm-mets-er-pr-neg-qqvt

Preclinical recommendations — forward-looking horizon scan

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Preclinical recommendations

A forward-looking horizon scan of candidate drugs, compounds, and treatment strategies that are earlier than clinical development, or not yet developed, but plausibly target this patient's stated features. These are research directions, not enrollable options or treatment recommendations.

1. PI3K/mTOR inhibition plus PD-1 blockade, selected by pS6-high uLMS (combination)

Target: High-grade uterine leiomyosarcoma histology **Stage:** Early translational **Evidence strength:** Moderate

Best-matched biology in the set, on one humanized PDX from one group. The drugs are not the hard part; the pS6 cutoff and a willing sponsor are.

Ranks first because it is aimed at the microenvironment this histology actually has: an M2-dominant myeloid compartment with exhausted CD8 T cells, which is also why pembrolizumab alone does nothing here. It is the only published intervention that reversed that phenotype in a humanized uLMS patient-derived xenograft and produced partial and complete responses alongside anti-PD-1, where anti-PD-1 by itself produced none. The selection marker is one IHC slide. What keeps the claim narrow: both drugs are already clinical in other diseases, so only the pairing and the pS6 cutoff sit at the preclinical boundary, and her pS6 has never been measured.

Mechanism. High phospho-S6 marks PI3K/mTOR overactivation in leiomyosarcoma and tracks with lymphocyte-depleted tumours. Blocking the pathway pushes macrophages off the pro-tumour phenotype and raises antigen presentation, which gives PD-1 blockade something to act on in a tumour that otherwise ignores it.

Translatability. High for the histology: a humanized uLMS patient-derived xenograft, the right immune phenotype, and a pS6 IHC readout any pathology lab could run.

Developability. Closest to reach in this set. Both halves are approved or late-stage elsewhere, so what is missing is a sponsor and a pS6-selected sarcoma protocol.

Counter-productive MoA. Moderate (Trabectedin already depletes tumour macrophages, so the two could cancel rather than combine)

Key risks.

- Only the pairing and the pS6 cutoff are preclinical; both drugs have been in patients for other diseases, so the novelty is narrower than it looks
- pS6 has not been measured on her tissue, so the selection step the whole result depends on has not happened
- A humanized mouse immune system is reconstituted rather than native, and it flatters how well a mouse can stand in for a human myeloid compartment
- Pneumonitis from PD-1 blockade and hyperglycaemia from PI3K/mTOR inhibition are an awkward fit for bilateral pulmonary disease at 8,500 ft
- One model, one group. Nothing has been tested alongside trabectedin, which depletes the same macrophage compartment

Open questions.

- Is her tumour pS6-high? That is a single IHC slide and nobody has run it
- Would the combination survive alongside trabectedin, or does trabectedin's macrophage depletion take the same lever first?
- Does the humanized-PDX response reproduce in a second model or a second laboratory?

Key references. pmid:38711203; pmid:40948800

2. CD70-directed antibody-drug conjugate (anti-CD70-MMAF) (ADC)

Target: High-grade uterine leiomyosarcoma histology **Stage:** In vivo (animal) **Evidence strength:** Emerging

An antigen handle in a tumour with almost no druggable mutations, resting on one therapeutic paper and an unexplained mismatch between tissue staining and cell-line detection.

uLMS is a disease of loss rather than of activating drivers, so a surface antigen is one of the few handles that does not require a mutation to exist. CD70 came out of an unbiased membrane-proteomics screen, sat above normal myometrium in 19 of 21 uLMS specimens, and the conjugate shrank both a cell-line xenograft and a patient-derived xenograft. It ranks second rather than first on how thin the therapeutic evidence is: one 2021 paper, one group. The antigen call is also less settled than the 90% tissue figure suggests, since only one of three uLMS lines was CD70-positive by western blot and nobody has explained the discordance. Her CD70 status is unmeasured and is not on the case's biomarker survey.

Mechanism. Membrane proteomics of uLMS lines against normal myometrium picked CD70 out as a differentially expressed surface antigen. An anti-CD70 antibody carrying a monomethyl auristatin F payload internalises after binding and delivers a mitotic poison to the cell that displayed the antigen.

Translatability. Moderate: a uLMS patient-derived xenograft and 19 of 21 clinical specimens, offset by antigen-detection discordance across the cell lines and no bystander payload.

Developability. The conjugate is academic, but CD70 binders exist elsewhere, so the bottleneck is a sponsor willing to run sarcoma and a CD70 IHC assay.

Counter-productive MoA. Moderate (Killing CD70-positive activated T cells could suppress the antitumour immunity other approaches here depend on)

Key risks.

- The therapeutic result is a single 2021 paper from a single group
- Only one of three uLMS lines was CD70-positive by western blot despite 90% positivity in tissue, and that gap is unexplained
- MMAF is non-cleavable and has no bystander effect, a poor pairing with an antigen that may be patchy
- CD70 sits on activated T cells, B cells and dendritic cells, so immune-compartment toxicity is a live question
- Her CD70 status is unmeasured and is not on the biomarker survey

Open questions.

- Does her tumour express CD70, and at what density? No assay has been run
- Why did western blot and tissue IHC disagree so badly, and which one predicts response?
- Would a cleavable, bystander-capable payload do better against a patchy antigen?

Key references. pmid:32822640; pmid:34656365

3. FANCM-RMI interaction inhibitors (cell-active cyclic peptides) for ALT-positive tumours (other)

Target: Unmeasured molecular landscape (the actionable gap) **Stage:** In vitro **Evidence strength:** Moderate

The strongest mechanism in this set and the least developed chemistry. Worth carrying only if ATRX or C-circle testing shows this tumour maintains telomeres by ALT.

ATRX loss and ALT are common in uLMS, and the ALT vulnerability the field reached for first, ATR inhibition, did not survive replication. This case file logs that failure on the record rather than assuming the branch closed. FANCM is the ALT dependency with the better record: two groups published it independently in the same 2019 issue of Nature Communications, building on a 2017 prior from a third group, and 2025 brought the first cell-active chemical matter. It earns rank 3 on mechanism, not on maturity. Everything downstream is contingent on ATRX sequencing or a C-circle assay, neither of which has been done, and the chemistry has never been given to an animal.

Mechanism. FANCM keeps telomeric replication stress in ALT cells below the level that kills them, using its translocase activity and its grip on the BLM-TOP3A-RMI complex. Remove it and ALT telomeres run into fork collapse and telomere dysfunction, while telomerase-positive cells carry on. Cyclic peptides that occupy the FANCM pocket on RMI1/2 do the same thing pharmacologically.

Translatability. Low: all cellular activity is in ALT-positive osteosarcoma lines, no leiomyosarcoma model has been tested, and her ALT status is unmeasured.

Developability. Earliest in the top group. A nanomolar cyclic peptide bolted to a cell-penetrating carrier, active in culture and nowhere else, with no pharmacokinetics reported.

Counter-productive MoA. Low (FANCM loss raises ALT activity before it kills; partial inhibition could aid telomere maintenance)

Key risks.

- ALT and ATRX status are unmeasured here, so the selection biomarker for the entire idea is missing
- FANCM restrains replication stress away from telomeres too; systemic inhibition could destabilise normal proliferating tissue
- Every cellular result is in osteosarcoma lines. No leiomyosarcoma model has been tried
- The ATR story is the cautionary one: an ALT-selectivity claim can look solid and then fail on isogenic re-testing

Open questions.

- Is this tumour ALT-positive? ATRX sequencing or a C-circle assay would answer it and neither has been ordered
- Does FANCM dependence hold in an ALT-positive leiomyosarcoma model, or only in osteosarcoma lines?
- Can a peptide this size be made to behave in an animal, or does a small molecule have to come first?

Key references. pmid:31138797; pmid:31138795; pmid:28673972; pmid:40479515; pmid:41236382; pmid:41436729

4. SUV39H2 inhibitor OTS186935 combined with a PARP inhibitor (combination)

Target: High-grade uterine leiomyosarcoma histology **Stage:** In vivo (animal) **Evidence strength:** Emerging

The best disease match in this scan and the thinnest support for it. One paper, one group, and a synergy index the authors themselves flag for validation.

Two things are true about this row and they pull in opposite directions. It has the best disease match in the file: SUV39H2 was higher in uLMS tissue than in myometrium or leiomyoma, the work was done in uLMS lines rather than borrowed from another histology, and the combination beat either agent alone in a uLMS xenograft. It also has the thinnest evidence base in the file: one paper, one laboratory, two cell lines, synergy in only one of them, and a combination index of 0.88 that sits barely inside the synergy range and could move with assay conditions. The authors say as much and ask for validation before anyone reads it clinically. The appeal is that it would create a repair deficit rather than wait to find one, which matters when her HR status is untested; the catch is that the olaparib half may still depend on that same untested status.

Mechanism. SUV39H2 methylates H3K9 and helps bring phosphorylated H2AX to double-strand breaks. Inhibiting it with OTS186935 blunts gamma-H2AX accumulation and degrades break repair, which is the state in which PARP inhibition becomes lethal rather than merely cytostatic.

Translatability. Moderate for the histology, low for the finding: real uLMS lines and a uLMS xenograft, but one marginal synergy result in one of two lines.

Developability. OTS186935 is a probe: no tolerability, no formulation, no sponsor. The PARP partner is off the shelf, so the whole gap sits on SUV39H2.

Counter-productive MoA. Low (Losing H3K9 methylation could derepress programmes unrelated to repair, with an unpredictable direction of effect)

Key risks.

- One laboratory, one paper, and the synergy appeared in only one of two cell lines
- A combination index of 0.88 is barely inside the synergy range and could move with assay conditions
- The olaparib contribution may hinge on HR status, which is untested here
- No toxicology at all for SUV39H2 inhibition, and H3K9 methylation does plenty outside DNA repair
- SK-UT-1 and SK-LMS-1 are the standard uLMS lines but both carry long-running identity and provenance questions

Open questions.

- Does the SK-UT-1 synergy hold in a second laboratory, and in a line without provenance questions?
- Is the olaparib half doing anything in an HR-proficient tumour, which nobody here has excluded?
- What does inhibiting SUV39H2 do to a whole animal over weeks? No toxicology exists

Key references. pmid:41402789

5. Ferroptosis blockade as anthracycline cardioprotection (ferrostatin-1,

MitoTEMPO, GPX4 induction) (small molecule)

Target: High-grade uterine leiomyosarcoma histology **Stage:** In vivo (animal) **Evidence strength:** Moderate

Best-replicated row in the scan, and the least actionable: no clinical-grade ferroptosis inhibitor exists, and nobody has ruled out the same protection shielding the tumour.

The only row here that touches what happens to her in the next few months. She is under 40, systemic-therapy naive, and about to start doxorubicin with a maintenance phase, so late cardiac injury is a decades-long problem rather than a treatment-window one. The evidence is unusually good for a preclinical row: two independent groups, genetic and pharmacologic confirmation on both sides, and a head-to-head in which ferrostatin-1 outperformed dexrazoxane. It is capped at moderate for two reasons that have nothing to do with the biology. No ferroptosis inhibitor exists in clinical-grade form, so nothing here is orderable. And nobody has asked whether the same protection shields the tumour, which is the question that would decide whether this helps her or quietly costs her the anthracycline's effect. It sits at 5 because it is supportive care rather than disease-directed. Baseline echocardiography is the step that is actually available today.

Mechanism. Doxorubicin degrades haem through Nrf2-driven Hmox1 induction, releasing free iron that piles up in cardiomyocyte mitochondria and drives lipid peroxidation. Ferrostatin-1, mitochondrial iron chelation, MitoTEMPO scavenging or raised GPX4 each prevent the cardiomyocyte death that dexrazoxane only partly prevents.

Translatability. Moderate: mouse cardiomyopathy models with genetic and pharmacologic agreement across two groups, but nothing tested under chronic hypoxic adaptation at 8,500 ft.

Developability. Ferrostatin-1 and MitoTEMPO are bench reagents with no human formulation. A clinical-grade radical-trapping antioxidant would come first, then a cardiology-endpoint trial, not a sarcoma one.

Counter-productive MoA. High (Blocking ferroptosis may protect the tumour from doxorubicin as well as the heart. Untested)

Key risks.

- Nobody has tested whether the same protection extends to the tumour. If ferroptosis contributes to doxorubicin's antitumour effect, this could blunt the treatment it is meant to enable
- No ferroptosis inhibitor exists in clinical-grade form, so nothing here can be ordered
- Chronic hypoxic adaptation at 8,500 ft shifts iron handling and haemoglobin, and no model has looked at anthracycline ferroptosis under those conditions
- Dexrazoxane already has human data and would not be displaced by a mouse result

Open questions.

- Does ferroptosis inhibition protect the tumour as well as the heart? Until someone answers that, this cannot move
- Does hypoxic adaptation at altitude change anthracycline iron handling at all?
- Would a clinical-grade radical-trapping antioxidant reproduce ferrostatin-1's advantage over dexrazoxane?

Key references. pmid:30692261; pmid:32376803; pmid:38232458

6. Anti-MARCO antibody reprogramming of suppressive tumour-associated

macrophages (antibody)

Target: Bilateral pulmonary metastases, infiltrative and possibly inflammatory **Stage:** In vivo (animal) **Evidence strength:** Moderate

Good mechanism, wrong tumour type so far. Reprogramming macrophages rather than deleting them is complementary to trabectedin in theory and untested in sarcoma in practice.

Reprogramming macrophages instead of removing them is a different lever from trabectedin's, which is the argument for carrying it in a case where trabectedin is about to be given. Three papers, two independent groups, and a second effector arm through NK cells give the mechanism more weight than most rows below it. What it does not have is any sarcoma data: every efficacy result is in carcinoma and melanoma models, and MARCO expression in uterine leiomyosarcoma has never been measured by anyone. The case for reading it across rests on the two microenvironments looking alike, which is a weak argument on its own. MARCO also sits on alveolar macrophages, a specific problem for someone with bilateral pulmonary metastases and reduced reserve at altitude.

Mechanism. MARCO is a scavenger receptor marking a suppressive macrophage subset. An antibody against it works through the inhibitory Fc receptor FcgammaRIIB to flip those macrophages toward a pro-inflammatory state rather than deleting them, and in parallel licenses NK-cell killing of the tumour.

Translatability. Low: carcinoma and melanoma models only, no sarcoma has been tested, and nobody has measured whether uLMS macrophages carry MARCO at all.

Developability. Research-grade antibodies only. No humanized candidate, no tolerability package, and the FcgammaRIIB dependence makes the Fc engineering harder than usual.

Counter-productive MoA. Moderate (Trabectedin depletes the macrophages this antibody needs alive in order to reprogram them)

Key risks.

- MARCO is also on alveolar macrophages, so a myeloid-directed antibody in someone with bilateral pulmonary disease at altitude carries a specific lung-toxicity concern
- MARCO expression in uterine leiomyosarcoma has not been measured by anyone
- Trabectedin is already working on the same compartment, so the two could be redundant rather than additive
- No sarcoma model has been tested, and sarcoma myeloid biology is not interchangeable with carcinoma myeloid biology

Open questions.

- Do uLMS macrophages express MARCO? A single stain on archival tissue would answer it
- Are trabectedin and MARCO reprogramming additive or redundant?
- How much of the lung's resident macrophage pool would an anti-MARCO antibody hit?

Key references. pmid:27210762; pmid:33293426; pmid:33229588; pmid:40948800

7. Arginine deprivation combined with BCL-XL blockade in ASS1-deficient

sarcoma (combination)

Target: Unmeasured molecular landscape (the actionable gap) **Stage:** In vivo (animal) **Evidence strength:** Emerging

The clearest development path in the lower half and the worst toxicity fit: BCL-XL platelet loss on top of doxorubicin and trabectedin is a hard sell.

ASS1 loss is recurrent in sarcoma and is not on this case's biomarker survey, so it is an unmeasured vulnerability sitting inside the stated molecular gap. What makes the work interesting is that it explains why arginine deprivation underperformed in patients rather than simply trying it again. Both components have been given to humans in other settings, so this row is closer to a protocol than most of the scan. Two things hold it here. Essentially one laboratory produced both cited papers, and BCL-XL inhibition brings dose-limiting thrombocytopenia that stacks badly on doxorubicin and trabectedin, which is precisely the regimen she is about to start. ASS1 itself is unmeasured.

Mechanism. Sarcomas that silence argininosuccinate synthetase 1 cannot make their own arginine and depend on the extracellular supply. Cutting that supply arrests them without killing them, because BCL-XL holds BAX and BAK shut. A BCL-XL inhibitor releases them and turns the arrest into cell death.

Translatability. Moderate: ASS1-deficient sarcoma lines with in-vivo efficacy modelling, but one group behind both papers and no leiomyosarcoma-specific result.

Developability. Both halves have been in patients elsewhere, so this needs a trial rather than chemistry. Platelet toxicity has stalled the BCL-XL class for years.

Counter-productive MoA. Moderate (Thrombocytopenia could force dose reductions in the chemotherapy this is meant to supplement)

Key risks.

- BCL-XL inhibition causes dose-limiting thrombocytopenia, which stacks badly on doxorubicin and trabectedin myelosuppression
- ASS1 status is unmeasured here and is not on the biomarker survey
- Arginine deprivation monotherapy has already disappointed in sarcoma, so the combination inherits a discouraging clinical prior
- One group generated most of this evidence

Open questions.

- Is this tumour ASS1-deficient? One IHC stain would tell, and it is not on the survey
- Can BCL-XL be inhibited without the platelet cost, for example with a platelet-sparing degrader?
- Does the combination hold up outside the group that generated it?

Key references. pmid:39898973; pmid:27735949

8. IL1R2 decoy-receptor blockade to release antitumour neutrophils in sarcoma (conceptual strategy)

Target: Bilateral pulmonary metastases, infiltrative and possibly inflammatory **Stage:** In vivo (animal) **Evidence strength:** Emerging

A genuinely surprising result with nothing to give a patient: the intervention is a germline knockout, and no IL1R2-blocking molecule exists in any programme.

Her pulmonary lesions were read as unusually infiltrative and possibly inflammatory, and this is the one row aimed at that reading rather than at the tumour cell. The effect was selective for sarcoma across the four tumour types tested, which is unusual enough to be worth carrying for this histology, and an IL1R2-deficiency signature tracked with better outcomes in human sarcoma. Then the problems. The intervention in the paper is a germline knockout, not a drug: no IL1R2-blocking molecule exists anywhere, and it is not obvious an antibody would reproduce lifelong deficiency in an adult with established metastases. One paper, one laboratory, no replication. Blocking a decoy receptor in order to increase its ligand's signalling is also an unusual pharmacological ask. And pushing more neutrophils into lungs that already carry bilateral infiltrative disease could make her worse before it did anything useful.

Mechanism. IL1R2 is a decoy receptor that soaks up IL-1 and damps its signalling. Removing it in mice raises IL-1-driven emergency granulopoiesis and floods the tumour with neutrophils whose transcriptional programme shifts away from pro-tumour functions toward activation and maturation.

Translatability. Low: mouse sarcoma models with a correlative human signature, nothing in leiomyosarcoma, and the intervention was genetic deletion rather than a drug.

Developability. There is no molecule at all. A tool antibody and a conditional, adult-onset deletion experiment would both have to come before anyone drew a protocol.

Counter-productive MoA. High (Neutrophils are usually pro-metastatic in lung; recruiting more could accelerate the pulmonary disease)

Key risks.

- Genetic deficiency from birth is not the same thing as blocking a receptor in an adult with established metastases
- One laboratory, one paper, no replication
- Recruiting neutrophils into lungs already carrying bilateral infiltrative disease could worsen symptoms before it helped
- The broader literature mostly reports neutrophils as pro-metastatic in lung, so this cuts against the prevailing direction
- The human data is a correlative signature, not a treated cohort, and nothing has been tested in leiomyosarcoma

Open questions.

- Would deleting IL1R2 in an adult with established disease do what deleting it from birth does?
- Does any of this hold in leiomyosarcoma, or only in the transplantable models tested?
- Can a decoy receptor be blocked pharmacologically without flooding the system with IL-1 signalling?

Key references. pmid:41824510

9. RAD52 inhibition (6-OH-dopa, RPA:RAD52 interaction inhibitors) in

BRCA2-deleted tumours (small molecule)

Target: Unmeasured molecular landscape (the actionable gap) **Stage:** In vitro **Evidence strength:** Emerging

Three independent chemotypes against one target, and none of them has left cell culture in ten years. Behind PARP inhibition in any HRD-confirmed queue.

The HRD subset of uLMS arises largely through BRCA2 homozygous deletion, meaning no protein rather than a hypomorph, which is the cleanest genetic background a RAD52 dependency could ask for. Three groups and three chemically distinct chemotypes point at the same target, which is more independent support than several rows above it. It sits at 9 anyway. Nothing has moved past cell culture in a decade of published work, which is itself informative about how hard the target is; 6-OH-dopa is a redox-active catechol whose promiscuity makes any selectivity claim hard to attribute; and if HRD is confirmed here, PARP inhibition is far better developed and would come first. This only becomes interesting after that fails. Her HR status and BRCA2 copy number are untested.

Mechanism. RAD52 runs a backup single-strand annealing route that BRCA-deficient cells lean on once homologous recombination is gone. Molecules that collapse RAD52's undecamer rings into dimers, or block its contact with RPA, take that backup away and are selectively lethal in BRCA-deficient cells.

Translatability. Low: cell lines and patient leukaemia cells only, no in-vivo work in a decade, and no leiomyosarcoma model has been tested.

Developability. All three chemotypes are probes with no in-vivo efficacy and no tolerability package. A real medicinal-chemistry campaign has not happened.

Counter-productive MoA. Low (Distinct lesion from PARP trapping, so little mechanism-level conflict with the goal)

Key risks.

- HR status and BRCA2 copy number are untested here, so the selection biomarker does not exist yet
- 6-OH-dopa's redox activity makes clean target attribution hard and any cellular selectivity suspect
- No leiomyosarcoma model has been tested
- If HRD is confirmed, PARP inhibition is far better developed and comes first; this only matters after that fails

Open questions.

- Is BRCA2 deleted here? Copy-number-aware sequencing would answer it; a mutation-only panel would not
- Can any RAD52 inhibitor be made to work in an animal?
- Does RAD52 dependence survive in a BRCA2-null leiomyosarcoma model rather than in the leukaemia and isogenic lines tested?

Key references. pmid:26548611; pmid:26784987; pmid:33784323

10. PLOD2 / lysyl hydroxylase 2 inhibition to reverse hypoxic collagen cross-linking in sarcoma (conceptual strategy)

Target: Bilateral pulmonary metastases, infiltrative and possibly inflammatory **Stage:** In vivo (animal) **Evidence**

strength: Emerging

Attractive concept for these lungs, sourced from an unrefereed preprint, with the only named drug ruled out as a tool by its own follow-up literature.

The hypoxia-PLOD2 mechanism is already in this case's preclinical evidence as biology. What is new here is only the attempt to intervene on it, and that attempt carries two problems that keep it near the bottom. The primary source is a bioRxiv preprint that had not been refereed at the time of writing, and the only pharmacology reported is minoxidil, which a published cautionary note says cannot serve as a lysyl-hydroxylase tool in vivo. The concept does fit these lungs, where a metastasis-promoting matrix and an immune-cold tumour describe what appears to be happening, and hypoxia is plausible in someone living at 8,500 ft, though nobody has studied that interaction. The direction of effect is not even settled: loosening collagen cross-links could help dissemination rather than restrain it. Systemic minoxidil also brings fluid retention, tachycardia and pericardial effusion, which pairs poorly with an anthracycline.

Mechanism. Hypoxia drives HIF-1alpha to induce PLOD2, which hydroxylates collagen lysines and organises the matrix into the aligned, stiff configuration that supports sarcoma dissemination. Its collagen VI product also pushes CD8 T cells toward dysfunction, so suppressing PLOD2 acts on the metastatic scaffold and the immune exclusion together.

Translatability. Low: undifferentiated pleomorphic sarcoma only, no leiomyosarcoma model, and the primary source has not been through peer review.

Developability. No selective LH2 inhibitor exists. Minoxidil suppresses PLOD transcription broadly instead of inhibiting the enzyme, and a published note rules it out in vivo.

Counter-productive MoA. High (Loosening the collagen scaffold could ease tumour dissemination instead of restricting it. Direction unsettled)

Key risks.

- The primary source is an unrefereed preprint
- The only reported pharmacology is minoxidil, and a published cautionary note says it cannot serve as an LH2 tool in vivo
- Systemic minoxidil causes fluid retention, tachycardia and pericardial effusion, a poor pairing with an anthracycline
- Loosening collagen cross-links could aid dissemination rather than restrain it; the direction of effect is unsettled
- All efficacy data is in undifferentiated pleomorphic sarcoma, not leiomyosarcoma

Open questions.

- Does the preprint hold up in peer review?
- Is there any selective LH2 inhibitor to be found, or is the enzyme not tractable?
- Does loosening the matrix help or hurt? Nobody has settled the direction

Key references. biorxiv:10.1101/2024.06.28.601055; pmid:33020194

11. Collateral-lethality mapping of the 13q homozygous-deletion footprint in uLMS (conceptual strategy)

Target: Unmeasured molecular landscape (the actionable gap) **Stage:** Conceptual **Evidence strength:** Speculative

A direction, not a candidate: no compound, no model, no analysis yet. Its near-term value is the argument for copy-number-aware sequencing over a mutation-only panel.

Last because there is nothing to give anyone: no compound, no programme, and an empty model list. It is logged so the direction stays visible rather than disappearing. uLMS is defined by loss rather than gain, and 13q is one of its recurrent losses, which makes it the natural place to look for a dependency created by the deletion itself. The CYCLOPS work showed that partial copy loss of core essential genes produces exactly this kind of liability, and collateral-lethality precedents from other tumour types show it can occasionally be drugged. Nobody has run the analysis for the leiomyosarcoma 13q footprint. Its only near-term consequence is an argument for copy-number-aware sequencing rather than a panel that reports mutations alone, which is the same workup the top-ranked row and several others here are waiting on.

Mechanism. Homozygous deletions rarely take only the gene they are named for. A 13q event that removes BRCA2, and often RB1 with it, takes the neighbouring passengers too. Where a passenger is a redundant member of an essential gene family, or a core essential gene left at reduced dosage, the surviving paralogue or residual copy becomes a dependency normal cells do not share.

Translatability. Low: no model system exists, and the reasoning is transposed from pancreatic and glioma precedents rather than tested in leiomyosarcoma.

Developability. Nothing to develop yet. The work is computational first: segment 13q across uLMS cohorts, intersect with dependency data, then ask whether anything is druggable.

Key risks.

- Entirely untested in leiomyosarcoma; the reasoning comes from pancreatic and glioma precedents
- Deletion boundaries differ between patients, so any vulnerability would be per-tumour rather than per-disease and would need her own copy-number data
- CYCLOPS dependencies are usually dosage effects on core essential genes, which makes the therapeutic window narrow by construction
- No sequencing has been done here, so even the first analytical step is blocked

Open questions.

- Where exactly does her 13q deletion start and stop? A mutation-only panel will not say
- Does any passenger in the uLMS 13q footprint have a druggable paralogue?
- Is the therapeutic window for a CYCLOPS dependency ever wide enough to matter clinically?

Key references. pmid:22901813; pmid:37143137; pmid:32299819